

CITYTWIN · CALCULATION

CityTwin sizing calculations

Open Hardware Portfolio · amishchadha.com

DOCUMENT CONTROL

Document ID	CTW-CAL-001	Version	0.1
Status	Draft	Date	2026-09-25
Author	Amish Chadha	License	CERN-OHL-S-2.0
Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/citytwin		
Source	docs/04-calcs/01-sizing.md @ c1de765		

REVISION HISTORY

Version	Date	Author	Change
0.1	2026-09-25	Amish Chadha	First issue for TRL 3 (radio load, storage and export, freshness and button response, legibility and shading, accessibility geometry, head and cabinet thermal with a heater study, power, wind and mass, backup, cost) with a status for every requirement

CityTwin sizing calculations

On paper, CityTwin meets 14 of its 20 requirements (7 by calculation, 7 by design), has 2 at risk and misses 4. **R10 is not met:** a button press takes 19.8 s to show a new layer, because the color e-paper needs about 19 s for a full refresh. **R16 is not met** against `budget_usd`: the kiosk parts cost \$739.00 against \$600, though they fit the \$750 recommended in the TRL 2 review, which awaits Amish. **R18 is not met in the worst case:** TwinKit's backup pack rides through 2.40 h when new but only 1.63 h when aged and at 0 °C, against 2 h. **R20 is not met:** in a street kiosk the panel would reach about 79 °C in low sun at 45 °C air and needs about 13.5 W of heat at -20 °C. R1 (ingest of every node type) and R12 (pilot climate, with no margin at 40 °C) are at risk. The radio and data side has ample margin: the 46-node reference neighborhood sends 5,616 uplinks a day and loses 0.61 % to collisions and downlink blanking. Several TRL 2 figures were corrected (last section). Every number in this note is printed by `docs/04-calcs/sizing.py`; the tag in brackets, for example [A2], is the line of that script's output that carries it.

Safety: These are first-principles screening estimates for a paper proof of concept. They do not replace the electrician's check of the mains installation, a structural design of the footing and anchors to the local code, or a privacy impact assessment. See CTW-PRC-001, Safety.

■ Scope and method

The note checks every requirement in CTW-REQ-001 v0.3 against the design in CTW-PRC-001 v0.3, the decisions in CTW-DDR-001 and the parametric model `cad/src/model.py`. The script imports the model's `PARAMS`, derived dimensions and part volumes, so the head, window, hood, cabinet, post and rail dimensions used here are the ones in the STEP files and in drawing CTW-DWG-001. It also reads `bom/bom.csv` and `budget_usd` in `project.yaml`, and writes the requirement table to `docs/04-calcs/results.csv`. Run it from the repo root with `python docs/04-calcs/sizing.py`.

The design case is the reference neighborhood of CTW-DDR-001 D9 on one TwinKit gateway in the kiosk cabinet (D4), a pilot kiosk on a sheltered site (D3), and a street kiosk case for R20.

■ Assumptions

Table 1. Main assumptions. Sibling figures are from those repos' TRL 3 calculation notes.

Area	Assumption	Basis
Node traffic	CurbCount 8 x 96 uplinks a day, 14 B; CrossSafe 4 x 24, 20 B; LoadZone 12 x 124, 12 B; HeatMap Node 6 x 96, 20 B; FloodGauge 4 x 96, 20 B (60 an hour in events); AirStreet 6 x 288, 20 B; NoiseMap 6 x 96, 22 B	CBC, LDZ, HMN, FLG, AST and NSM CAL-001; the CrossSafe rate is a CityTwin assumption
Radio	13 B of LoRaWAN overhead, 125 kHz, CR 4/5, 8-symbol preamble; every node at SF9; pure ALOHA over 8 channels; no capture effect	TWK-CAL-001 method, conservative
Downlinks	LoadZone sign option on all 6 two-puck bays, 200 downlinks and 33 s of transmit a day each; the gateway cannot receive while it transmits	LDZ-CAL-001 [H5]
Peak hour	FloodGauge in event mode on all 4 gauges; LoadZone at 3 times its daily mean	Assumed storm and rush hour together
Pulled data	PotholeLog: 3 buses, 112 kB of summaries a bus-day, at most 6,000 published segment rows a day; DockHub: 24 dock-hour rows a day	PHL-PRC-001 v0.3; D11
Storage	90 B per stored value, uncompressed; values per record: CurbCount 7, CrossSafe 4, LoadZone 5, HeatMap 11, FloodGauge 6, AirStreet 10, NoiseMap 22, pulled rows 6; 56 GB free on the card, shared with TwinKit's own 4.73 GB a year	TWK-CAL-001; payload descriptions in the sibling notes
Timing	Network 4.7 s worst (TWK-CAL-001); layer render 30 s; kiosk redraw every 10 min; 1600 x 1200 frame at 4 bits per pixel over SPI at 10 MHz; PNG at 20 % of raw over 80 Mbit/s Ethernet; full refresh 19 s	Waveshare manual for the refresh; the rest assumed
Light	LED 100 lm/W; 40 % of the light reaches the screen	Assumed
Thermal	Outside film 10 W/m ² K, inside film 5 W/m ² K; solar absorptance 0.5 (grey paint); sun 600 W/m ² on a vertical face, 1,000 W/m ² on a horizontal one; window transmits 0.85, panel absorbs 0.7; 1 mm air gap and 6 mm polycarbonate in front of the panel; TwinKit processor 29.1 K above the air around it at light load (69.1 °C at 40 °C) and throttles at 85 °C; pack charges only at 0 to 45 °C	TWK-CAL-001 F2; handbook ranges; to be measured
Power	Controller 1.0 W; button LEDs 0.1 W; e-paper 0.5 W while refreshing; hood light 2 W for 12 h a night; TwinKit 6.41 W; 12 V supply 85 % efficient at light load; protection 0.2 W	TWK-CAL-001 D1; typical figures
Wind	35 m/s gust, air 1.25 kg/m ³ , load factor 1.5; drag coefficient 1.3 for the head, hood and cabinet, 2.0 for the square post, 1.2 for the antenna; all areas loaded at once; S355 post, 355 MPa yield; M16 anchor design tension 10 kN	Screening values; the local wind code and the anchor maker's data govern
Mass	Steel 7,850, aluminum 2,700, polycarbonate 1,200 kg/m ³ on the model volumes; bought-in items by estimate (display 1.2 kg, TwinKit 1.2 kg)	Model; estimates

■ A. Radio load and capacity (R1, R2)

- **Traffic.** The reference neighborhood of 46 LoRaWAN nodes sends 5,616 uplinks a day (TRL 2: 4,896), mainly because LoadZone's TRL 3 note sets 124 uplinks a day per puck, not 64 [A1, A2]. At SF9 the uplinks take 1,309 s a day, 1.52 % of one channel, or 0.189 % of each of 8 channels [A2].
- **Loss.** Collisions lose 0.38 % of uplinks. With the LoadZone sign option on all six bays, 1,200 downlinks take 198 s of gateway transmit a day, during which the gateway hears nothing: another 0.23 %, for 0.61 % in all [A3]. In a

storm hour with every FloodGauge in event mode and LoadZone at three times its mean, the gateway sees 582 uplinks an hour and loses 0.94 % plus the same blanking [A4].

- **Capacity (R2).** 10,000 records a day of 20 B at SF9 lose 0.71 %; one gateway takes up to 14,075 a day before loss reaches 1 %, consistent with TwinKit's 1.02 % at 14,400 [A5]. **R2 is met on paper.**
- **Pulled sources (R1).** PotholeLog uploads its daily summaries (3 buses x 112 kB) to the fleet operator's server, not to CityTwin, and DockHub syncs its logs over LTE-M to its own back end. Under D11 the gateway fetches the published PotholeLog segment files (at most 6,000 rows a day) and DockHub's 24 dock-hour rows a day by outbound HTTPS, so it still accepts no inbound connection [A6]. CrossSafe's radio runs LoRa point to point between the two sides of a crossing and has no LoRaWAN uplink defined; the 24 summaries a day assumed here depend on CrossSafe adding one. **R1 is at risk:** met on paper for six LoRaWAN types, with the PotholeLog and DockHub paths proposed but not agreed and the CrossSafe uplink undefined.

■ B. Storage and open export (R2, R6)

- **Store.** 87,936 values a day take 2.89 GB a year uncompressed. Two years of raw records (D6) are 5.78 GB and ten years of hourly aggregates 15.14 GB, 20.9 GB in all [B1]. With TwinKit's own two-year case the card holds 30.4 GB of 56 GB free [B2].
- **Export (R6).** Each day the export carries 1,104 node-hour rows, up to 6,000 road segment rows and 24 dock rows: 7,128 rows, 0.86 MB as CSV and 2.85 MB as GeoJSON, or 1.35 GB a year on the public host [B3]. PotholeLog segments dominate; the TRL 2 figure of 0.2 MB a day left them out. **R6 is met by design;** the sizes confirm that one-way pushes of this volume are trivial.

■ C. Freshness and button response (R8, R10)

- **Data age (R8).** For a 15 min node, the newest value can be 15 min old when the next uplink arrives, and the kiosk may then wait up to 10 min for its next redraw. With 4.7 s of network time, 30 s to render the layer, 0.79 s to move the image and 19 s to refresh, the worst age on the kiosk is 25.9 min; if one uplink is lost (0.38 % chance) it is 40.9 min. The web map shows data at most 15.6 min old [C1]. **R8 is met on paper,** with a margin of about 4 min.
- **Button (R10).** A press starts a 960 kB frame transfer (0.79 s) and a full refresh (19 s): 19.8 s against 5 s [C2]. Caching every layer on the controller removes only the transfer. **R10 is not met;** the LED ring acknowledges the press at once. Options are listed in `docs/REVIEW.md`.

■ D. Legibility, night light and shading (R9)

- **Text.** The panel has 150 pixels per inch, so 7 mm capitals are 41 px tall and subtend 16.0 arcmin at 1.5 m: 3.2 times the 5 arcmin of a 20/20 letter and 1.6 times the 10 arcmin of 20/40 [D1].
- **Night.** The 2 W hood light gives about 1,459 lx on the 0.0548 m² screen; 200 lx would need only 0.27 W [D2], so the light can be dimmed. The power budget below keeps the full 2 W.
- **Sun.** The hood reaches 140 mm beyond the head. With the sun facing the screen, its shadow covers 13 % of the window at 30°, 31 % at 45° and 63 % at 60° elevation [D3]. The hood shades the screen at midday but not in low morning or evening sun, which matters for R20 below.
- **R9 is met on paper:** reflective e-paper reads in sun by design, and the text size and night light have margin.

■ E. Accessibility geometry (R11)

The buttons sit at 1,120 mm, inside the 380 to 1,220 mm reach range. The head's leading edge runs from 1,040 to 1,734 mm, inside the 685 to 2,030 mm zone (27 to 80 in) where objects on posts may overhang no more than 305 mm (12 in) ([U.S. Access Board, protruding objects](#), checked 2026-09-25). The head overhangs the post by 180 mm sideways and the hood by 200 mm sideways and 190 mm forward. The cabinet bottom, at 300 mm, is below 685 mm and within cane reach [E1]. **R11 is met on paper** for the kiosk; the WCAG 2.1 AA web page is not verifiable at TRL 3. The screen center at 1.46 m is high for a seated reader; the web page and the QR code carry the same content.

■ F. Thermal: head, heater study and cabinet (R12, R20)

- **Pilot head (R12).** The head holds 1.12 W of electronics and loses heat from 0.786 m², a rise of 0.14 K in shade. At 40 °C air the panel sits at 40.1 °C, just above its 40 °C rating; the pilot's usable air limit is 39.9 °C [F1].
- **Street head in sun (R20).** With 600 W/m² of low sun on the face, the head air rises 9.2 K, and the panel, which absorbs 19.6 W through the window, sits 33.6 K above the air outside: about 79 °C at 45 °C air [F2]. Only orientation or shading of the face can fix this; the hood does not shade low sun.
- **Heater study (R20).** Holding the panel at 0 °C in -20 °C air takes 156 W if the whole head is heated, but only 13.5 W if the panel sits in a bay lined with 20 mm of foam and the heater warms only that bay [F3].
- **Cabinet.** The sealed cabinet holds 8.11 W (TwinKit 6.41 W, supply loss 1.50 W, protection 0.2 W). Its air rises 4.4 K in shade and 18.8 K in sun, when it absorbs 78 W [F4]. In the pilot case (40 °C shaded) the cabinet air is 44.4 °C, the TwinKit processor 73.5 °C against 85 °C, and the pack 0.6 K below its 45 °C charge limit. At 45 °C in shade the pack is 4.4 K over the limit; in sun the cabinet air reaches 63.8 °C and the processor 92.9 °C, above its throttle point [F5]. At -20 °C air the cabinet is at -15.6 °C, too cold to charge the pack [F6].
- **Status. R12 is at risk:** the pilot range is met with no margin at its top end for both the panel and the pack. **R20 is not met.**

■ G. Power (R14, R15)

- **Average (R15).** The head draws 2.12 W of 12 V DC (controller 1.0 W, hood light 1.0 W averaged over 12 h a night, button LEDs 0.1 W, e-paper 0.016 W). With TwinKit's 6.41 W and an 85 % supply, the kiosk takes 10.2 W from the mains: 0.246 kWh a day, 90 kWh a year [G1]. **R15 is met on paper.** A street kiosk with the panel heater running at -20 °C would take 26.1 W while heating [G2], above 15 W; how long it runs depends on the site's climate.
- **Supply (R14).** The 12 V peak is 26.7 W (TwinKit 20.6 W, controller boot 2.5 W, e-paper 1.5 W, hood 2 W, LEDs 0.1 W), or 40.2 W with the heater, within the 60 W supply. The mains current is 44 mA at 230 V [G3], far below the 6 A RCBO rating. **R14 is met by design** and the supply size is confirmed.

■ H. Wind, structure and mass (R13)

- **Loads.** At 35 m/s the dynamic pressure is 766 Pa. The head and hood take 348 N at 1.38 m, the cabinet 165 N at 0.53 m, the post 268 N at 0.88 m and the antenna 11 N at 2.08 m [H1]: 792 N and a base moment of 826 N·m, or 1,239 N·m with the 1.5 load factor [H2].
- **Post and weld.** The 100 x 100 x 4 mm section (47,268 mm³) is stressed to 26.2 MPa, a factor of 13.5 on yield [H2]. A 4 mm fillet weld all round the post base sees 32.9 MPa [H3].

- **Anchors and plate.** Each anchor on the tension side carries 1.92 kN, a factor of 5.2 on the assumed 10 kN [H5]. The 12 mm base plate sees 40 MPa [H6]. The head moves 0.65 mm under the unfactored gust [H7].
- **Mass.** The kiosk weighs 58.2 kg (post 21.4, base 15.5, cabinet with rails 10.7, head 4.1 kg) [H4]. The TRL 2 figure was about 55 kg.
- **R13 is met on paper.** The TRL 2 figure of 12 MPa left out the post's own drag and the load factor. Vehicle impact is not covered; the footing and anchors are site design.

■ I. Outage behavior (R18)

TwinKit's backup runs the gateway for 2.40 h with a new pack and 1.63 h with an aged pack at 0 °C, against the 2 h of R18. The head is not on the backup; the bistable e-paper keeps its last image and time stamp [I1]. **R18 is not met in the worst case.** A cold cabinet in winter (section F) would shorten the runtime further.

■ J. Cost (R16, R17)

All 18 BOM lines are priced. The kiosk (items 1 to 12, 14 and 15) costs \$739.00; the TwinKit gateway \$290.00 from TwinKit's TRL 3 BOM; together \$1,029.00 [J1]. Against `budget_usd` of \$600 the kiosk is \$139.00 over; against the \$750 recommended under D1, which awaits Amish, it is \$11.00 under [J2]. **R16 is not met** against the current budget. **R17**, redefined by D1 as a reported figure with the gateway costed in TwinKit, is met on paper.

■ Results

Table 2. Requirement status (also in `docs/04-calcs/results.csv`).

ID	Requirement	Value	Target	Status
R1	Ingest every node type	6 LoRaWAN types on paper; PotholeLog and DockHub by outbound pull (D11), not agreed; CrossSafe has no LoRaWAN uplink	All 9 types	At risk
R2	Capacity on one gateway	46 nodes, 5,616 a day, 0.38 % loss; 10,000 a day gives 0.71 %	50 nodes, 10,000 a day	Met on paper
R3	Counts and levels only	Schema per node type at the decoder	Out-of-schema fields rejected	Met by design
R4	Small-count suppression	Threshold 5 per hour (D6)	Fewer than 5 published as such	Met by design
R5	No movement traces	Segment per day; dock per hour	No tracks or card IDs	Met by design
R6	Open data export	7,128 rows a day, 0.86 MB CSV	Hourly CSV and GeoJSON, CC BY 4.0	Met by design
R7	One map	Layer per type, node health, as-of time	As stated	Met by design
R8	Data freshness	25.9 min worst on the kiosk	30 min	Met on paper (thin margin)
R9	Readable day and night	41 px, 16.0 arcmin; 1,459 lx at night	Sun and night at 1.5 m	Met on paper
R10	Button response	19.8 s	5 s	Not met
R11	Accessible kiosk	Buttons 1,120 mm; overhang 200 mm	380 to 1,220 mm; 305 mm; WCAG page	Met on paper (web page not verifiable at TRL 3)
R12	Pilot climate (redefined, D3)	Panel 40.1 °C and pack 44.4 °C at 40 °C air	Sheltered site, 0 to 40 °C air	At risk
R13	Wind	26.2 MPa with the 1.5 factor; 1.92 kN per anchor	35 m/s x 1.5, no yield	Met on paper
R14	Electrical safety	Mains only in the cabinet; 12 V SELV; peak 26.7 W of 60 W	As stated	Met by design
R15	Running power	10.2 W	15 W	Met on paper
R16	Kiosk parts cost	\$739.00	\$600 (budget_usd); \$750 recommended, awaiting Amish	Not met
R17	Cost with gateway (redefined, D1)	\$1,029.00	Reported; gateway costed in TwinKit	Met on paper (reported)
R18	Honest in an outage	2.40 h new, 1.63 h worst	2 h ride-through	Not met (worst case)
R19	Secure by default	Outbound push and pull only	No inbound connections	Met by design

ID	Requirement	Value	Target	Status
R20	Street climate (target after the pilot)	Panel about 79 °C in sun at 45 °C; heater 13.5 W at -20 °C	-20 to +45 °C air	Not met

Summary: 4 not met, 2 at risk, 7 met on paper, 7 met by design [K1]. The script counts R11 and R17 under "met on paper".

■ Changes from the TRL 2 estimates

Table 3. TRL 2 figures checked against this note.

Quantity	TRL 2 (CTW-PRC-001 v0.2)	TRL 3 (this note)	Result
Records a day	About 4,896	5,616	LoadZone rate from LDZ-CAL-001
Radio airtime	About 900 s a day, 1 % of a channel, 0.13 % over 8	1,309 s, 1.52 %, 0.189 %	Corrected
Radio loss	About 3 % (assumed)	0.61 % with downlink blanking	Calculated
Storage	About 350 MB a year	2.89 GB a year raw, uncompressed	TwinKit's conservative 90 B per value
Open export	About 1,100 rows, 0.2 MB a day	7,128 rows, 0.86 MB CSV a day	PotholeLog segments added
Worst data age	About 25 min	25.9 min	Stands
Button to new layer	About 20 s	19.8 s	Stands; R10 still not met
Kiosk and gateway power	About 9 W, 75 kWh a year	10.2 W, 90 kWh a year	Hood light for 12 h, supply at 85 %
Post stress	About 12 MPa	26.2 MPa with the 1.5 factor	Post drag and factor added; still met
Anchor tension	About 1 kN	1.92 kN	As above
Mass	About 55 kg	58.2 kg	From the model
Kiosk parts	About \$740	\$739.00	Stands
Kiosk and gateway	About \$1,025	\$1,029.00	TwinKit now \$290.00
Backup pack	Separate pack in the cabinet	Pack inside TwinKit's UPS module (TWK-DWG-001)	Model follows TwinKit